OFFICIAL ENTERPRISE FRAMEWORK DOCUMENT

Cloud Infrastructure & Operations Architecture

Comprehensive Standards for Multi-Cloud Engineering, Security, Networking, Disaster Recovery & Governance

Company: Series Tech Limited

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Primary HQ/Branch: Bangalore HQ & Multi-Region Teams

Owner: Cloud Engineering Department

Target Audience: Cloud Engineers, DevOps, Software Engineers & Security

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Note: This document provides complete operational standards, multi-cloud architectures (AWS/Azure), security controls, disaster recovery benchmarks, and FAQs optimized for semantic search and retrieval.

1. Executive Purpose & Scope

Purpose: This document describes the complete cloud infrastructure framework at Series Tech Limited, covering AWS services, Azure services, cloud security, storage, networking, monitoring, backup, disaster recovery, access permissions, and cloud engineer responsibilities. It is written so that any employee working with cloud infrastructure at Series Tech Limited can understand these practices fully without needing another document.

Scope: This document applies to all employees in the Cloud Engineering department of Series Tech Limited, based primarily in the Bangalore branch office, as well as DevOps and Software Development employees across all offices (Hyderabad, Bangalore, Pune, Chennai, Mumbai) who interact with cloud environments for client projects.

2. Introduction & Strategy

Series Tech Limited builds and operates cloud infrastructure for its clients using both Amazon Web Services (AWS) and Microsoft Azure, selecting the appropriate cloud provider based on client preference, existing infrastructure, and project requirements. The Cloud Engineering department of Series Tech Limited is responsible for designing, provisioning, securing, and monitoring this infrastructure across multi-tenant and dedicated client environments.

3. Multi-Cloud Ecosystem Architecture

3.1 AWS Ecosystem Stack

Series Tech Limited utilizes AWS Well-Architected Framework principles to construct resilient environments:

- **Compute Workloads:** Elastic Compute Cloud (EC2), Elastic Container Service (ECS), Elastic

3.2 Azure Ecosystem Stack

For clients integrated into the Microsoft ecosystem, equivalent enterprise services are deployed:

- **Compute Workloads:** Azure Virtual Machines, Azure App Services, Azure Kubernetes Service (AKS), Azure Functions.

Kubernetes Service (EKS), AWS Lambda serverless functions.

- **Managed Databases:** Amazon RDS (Aurora MySQL/PostgreSQL), Amazon DynamoDB (NoSQL), ElastiCache (Redis/Memcached).
- **Storage Services:** Amazon Simple Storage Service (S3), EBS block storage, EFS shared file storage.
- **Content Delivery & CDN:** Amazon CloudFront integrated with AWS WAF for fast, secure edge distribution.

- **Managed Databases:** Azure SQL Database, Azure Cosmos DB (NoSQL), Azure Database for PostgreSQL/MySQL.
- **Storage Services:** Azure Blob Storage (Hot, Cool, Archive tiers), Azure Managed Disks, Azure Files.
- **Content Delivery & Networking:** Azure CDN, Azure Front Door, Azure Application Gateway.

4. Cloud Security & Zero-Trust Governance

> **Shared Responsibility Model:** Cloud providers (AWS/Azure) secure physical facilities, hypervisors, and underlying hardware. Series Tech Limited is strictly responsible for secure resource configurations, identity and access management (IAM), data encryption at rest/transit, network perimeter filtering, and continuous threat monitoring.

Infrastructure-as-Code (IaC) Requirement: All infrastructure must be provisioned via standardized, version-controlled Terraform or AWS CloudFormation/Azure ARM templates containing built-in security guardrails. Manual console tweaks are strictly forbidden in production.

Access Control & Least Privilege

- **Need-to-Know Access:** Permissions are restricted to minimal required roles using Role-Based Access Control (RBAC).
- **Quarterly Access Reviews:** The Cloud Engineering Lead conducts formal quarterly audit reviews of all active accounts and keys.
- **Offboarding Compliance:** Immediate revocation of access keys, VPN profiles, and IAM roles upon employee project departure.
- **Contractors & Interns Policy:** Authorized contractors undergo standard IaC verification workflows. Interns are explicitly restricted from production cloud environments.

5. Storage Tiering & Network Segmentation Architecture

Storage Lifecycle Management

Storage is categorized into performance and cost tiers based on data lifecycle policies:

- **Standard Tier:** Frequently accessed application data, active databases, and daily operations.
- **Archival Tier:** Low-cost AWS S3 Glacier / Azure Blob Archive for long-term compliance backups.
- **Encryption Protocol:** Mandatory AES-256 (at rest) and TLS 1.3 (in transit) for all storage buckets/ disks.

Network Tiering & Isolation

Each client operates within dedicated Virtual Private Clouds (VPC / VNet) segmented into three subnets:

- **Public Subnet:** Load balancers (ALB), API gateways, and bastion/NAT gateways only.
- **Application Subnet:** Microservices, container clusters, worker nodes (No public IPs).
- **Database Subnet:** RDS/SQL instances strictly isolated with Security Groups / NACLs.

6. Monitoring, Backup & Disaster Recovery (DR)

Observability & Telemetry: Series Tech Limited deploys CloudWatch (AWS) and Azure Monitor/Log Analytics paired with centralized SIEM integrations. Telemetry tracks CPU/memory spikes, database connection counts, HTTP 5xx error thresholds, and anomalous IAM API requests. Alerts route directly to the designated on-call Cloud Engineer.

Backup & Disaster Recovery Matrix

Category	Standard Production Policy	Critical Enterprise Benchmark
Backup Frequency	Automated daily snapshot backups.	Point-in-Time Recovery (PITR) with continuous log archiving.
Retention Period	Standard retention for 30 days in secondary regions.	Up to 7 years in cold archival storage for regulatory compliance.
Recovery Time Objective (RTO)	Target system recovery within 4 hours.	Sub-15 minute automated failover for multi-region setups.
Recovery Point Objective (RPO)	Maximum acceptable data loss: 1 hour.	Near-zero RPO using multi-region database replication.
Disaster Recovery Testing	Simulated regional failover drill held **at least annually**.	Semi-annual coordinated failover verification with client sign-off.

7. Governance, Roles & Stakeholder Responsibilities

Role / Department	Core Infrastructure Responsibilities
Cloud Engineering Lead	Reviews IAM permissions quarterly, ensures annual DR failover testing, approves production IaC releases, enforces architectural guardrails.
Cloud Engineers	Provision infrastructure via IaC templates, configure monitoring/alerts, maintain backup automation, update templates following emergency hotfixes.
DevOps Department	Collaborates on CI/CD pipeline automation, co-approves production changes alongside Cloud Engineering Lead.
Cybersecurity Department	Conducts independent periodic security reviews, vulnerability scans, and compliance audits of cloud tenant environments.

8. Operational Policies & Deployment Workflows

Infrastructure Gatekeeper Policy: Any manual change executed during an active incident (Console/CLI) MUST be backported into the version-controlled IaC repository within 24 hours to prevent configuration drift.

Deployment & Validation Pipeline

1. **Drafting IaC:** Engineer develops Terraform/CloudFormation code locally using verified base templates.
2. **Automated Validation:** Submission via Internal Cloud Portal triggers automated static security scans (e.g., tfsec, Checkov) checking for open security groups, unencrypted storage, or missing tags.
3. **Standard Approval:** Cloud Engineering Lead reviews and approves change request after validation passes.
4. **Production Approval:** Requires dual sign-off from Cloud Engineering Lead AND DevOps Lead following deployment gates in the QA & DevOps guidelines.

9. Employee Operational Scenarios & Common Pitfalls

Real-World Engineering Scenarios

Scenario 1 (Hotfix Recovery): An engineer makes an emergency manual console fix during a severe outage. *Action:* Immediately post-incident, update the IaC template to reflect the change and commit to Git.

Scenario 2 (Offboarding): A developer transfers off Project RetailEdge. *Action:* The project manager and Cloud Lead execute immediate IAM credential revocation during offboarding.

Scenario 3 (RTO Revision): A banking client requests a stricter 15-minute RTO. *Action:* Cloud Engineering Lead re-architects database replication (e.g., Aurora Global Database) and updates DR playbooks.

Common Pitfalls to Avoid:

1. **Configuration Drift:** Making un-tracked manual console adjustments that get overwritten in the next IaC pipeline run.
2. **Orphaned Credentials:** Failing to revoke IAM keys when personnel leave project teams.
3. **Under-Provisioning Backup Retention:** Misconfiguring snapshot lifecycles relative to client SLA contracts.

10. Glossary & Key Definitions

Infrastructure-as-Code (IaC): The practice of managing and provisioning cloud infrastructure through machine-readable configuration files (e.g., Terraform, CloudFormation) rather than manual console actions.

Recovery Time Objective (RTO): The target duration within which a business process or system must be restored after a service disruption.

Recovery Point Objective (RPO): The maximum acceptable period of data loss measured in time prior to the disruption.

11. Frequently Asked Questions (FAQ Knowledge Base)

Q1: Which cloud providers does Series Tech Limited use?

Amazon Web Services (AWS) and Microsoft Azure.

Q2: How does Series Tech Limited decide between AWS and Azure for a project?

Based on client preference, existing client infrastructure, and specific technical project needs.

Q3: What is the shared responsibility model?

A framework where the cloud provider secures the underlying physical infrastructure while Series Tech Limited secures resource configuration, access permissions, data encryption, and threat monitoring.

Q4: Is data encrypted in Series Tech Limited cloud environments?

Yes, encryption is mandatory both at rest (AES-256) and in transit (TLS 1.3) for all client data.

Q5: How is cloud access granted?

Strictly on a need-to-know and least-privilege basis using Role-Based Access Control (RBAC).

Q6: How often are access permissions reviewed?

Quarterly, by the Cloud Engineering Lead.

Q7: What happens when an employee moves off a project?

Their cloud access permissions are revoked promptly as part of the project offboarding process.

Q8: How often is disaster recovery tested?

At least annually, through a simulated failover exercise coordinated by the Cloud Engineering department.

Q9: What is a recovery time objective (RTO)?

The target duration within which a system must be fully restored after a disruption, as agreed with the client.

Q10: What is a recovery point objective (RPO)?

The maximum acceptable amount of data loss measured in time following an unexpected outage.

Q11: How are cloud resources provisioned at Series Tech Limited?

Exclusively through version-controlled Infrastructure-as-Code (IaC) templates rather than manual console changes.

Q12: Who approves standard infrastructure changes?

The Cloud Engineering Lead, after automated security and syntax validation passes.

Q13: Who approves production environment changes?

Both the Cloud Engineering Lead and the DevOps Lead.

Q14: What monitoring does Series Tech Limited configure?

Continuous monitoring for CPU utilization, memory usage, API error rates, network spikes, and unusual access patterns.

Q15: How is backup frequency determined?

Based on data criticality, ranging from automated daily snapshots to continuous Point-in-Time Recovery log archiving.

Q16: How long are backups retained?

Typically 30 days for standard systems, with longer archival retention (up to 7 years) configured for critical compliance workloads.

Q17: Is backup restoration tested?

Yes, periodic restoration drills are conducted to verify backup integrity and restore procedures.

Q18: What storage tiers does Series Tech Limited use?

Standard storage for frequently accessed data and archival storage (e.g., S3 Glacier, Azure Archive) for long-term backups.

Q19: How is cloud networking structured?

Using isolated Virtual Private Clouds (VPC/VNet) divided into distinct subnets for Public/ALB, Application, and Database tiers.

Q20: What restricts access between subnets?

Security Groups and Network Access Control Lists (NACLs) configured according to least privilege rules.

Q21: Can interns access production cloud environments?

No, interns are strictly restricted from production cloud environments.

Q22: Can contractors provision cloud infrastructure?

Yes, if authorized within the Cloud Engineering team and subject to standard IaC validation and lead approvals.

Q23: What happens if an infrastructure-as-code template fails validation?

Deployment is automatically blocked by the pipeline until compliance and security errors are resolved.

Q24: Who is responsible for infrastructure security configuration?

Cloud Engineers, following the Shared Responsibility Model and baseline security templates.

Q25: What are AWS well-architected principles?

Framework pillars (Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, Sustainability) followed during design.

Q26: How are cloud alerts routed?

Automated routing directly alerts the designated on-call Cloud Engineer for immediate incident triage.

Q27: What happens during a disaster recovery test?

A controlled simulated regional outage where failover procedures, DNS cutover, and RTO/RPO targets are benchmarked.

Q28: Who supports periodic security reviews of cloud environments?

The Cybersecurity Department, in close coordination with Cloud Engineering.

Q29: What database services does Series Tech Limited use on AWS?

Managed relational databases (Amazon RDS Aurora MySQL/PostgreSQL) and NoSQL services (Amazon DynamoDB).

Q30: What database services does Series Tech Limited use on Azure?

Azure SQL Database, Azure Cosmos DB, and Azure Database for PostgreSQL/MySQL.

12. Document Revision History

Version	Date	Summary of Changes	Approved By
1.0	Initial Release	First publication of the Series Tech Limited Cloud Infrastructure document.	Head of Cloud Engineering
1.1	Annual Review	Updated backup retention guidance, multi-cloud architecture stack, and disaster recovery testing frequency.	Head of Cloud Engineering